

§1.532 — why $B \times A_1$ is the pullback of x along the projection

Freyd & Scedrov, §1.532 — and why $(B \times -)$ preserves covers

§1.532 needs the functor $B \times (-)$ to preserve covers. The key: the square $B \times A_1 \rightarrow B \times A_2 \rightarrow A_2 \leftarrow A_1$ is a pullback in any category with binary products, and in a pre-regular category covers are stable under pullback. So a cover x pulls back to the cover $1 \times x$.

§1. The pullback square

The following square commutes and is a pullback in any category with binary products.

$$\begin{array}{ccc} B \times A_1 & \xrightarrow{1 \times x} & B \times A_2 \\ \pi \downarrow & & \downarrow \pi \\ A_1 & \xrightarrow{x} & A_2 \end{array}$$

Figure 1: The pullback square. π is the projection onto the A -factor (i.e. `fst`, dropping B). Commutativity in diagram order: $(1 \times x)\pi = \pi x$ — both extract the A -component and then apply x .

The cospan whose pullback this computes — the two arrows into the shared corner A_2 :

$$A_1 \xrightarrow{x} A_2 \xleftarrow{\pi} B \times A_2.$$

§2. Why $B \times A_1$ is the pullback, not just a product

A pullback in **Set** is a subset of a product cut out by an equation. At first glance $B \times A_1$ looks like a plain product with no equation in sight. The resolution: **it is** that subset — the equation just happens to determine one coordinate completely, so the constrained set collapses back to a product.

Concretely, the pullback of the cospan above is:

$$P = \{(a_1, (b, a_2)) \in A_1 \times (B \times A_2) \mid x(a_1) = a_2\}.$$

The equation $x(a_1) = a_2$ **determines** a_2 from a_1 : once a_1 is chosen, a_2 is forced to be $x(a_1)$, carrying no extra information. The genuinely free data is the pair (b, a_1) . Hence

$$P \cong B \times A_1,$$

via the bijection $(a_1, (b, a_2)) \mapsto (b, a_1)$, with inverse $(b, a_1) \mapsto (a_1, (b, x(a_1)))$.

So $B \times A_1$ is not “a product **instead of** a pullback” — it **is** the pullback, written without the now-redundant coordinate a_2 .

Slogan. Pulling back along a projection just carries the constant factor through. $B \times A_2$ is B base-changed from 1 up to A_2 ; base-changing once more along $x : A_1 \rightarrow A_2$ gives B over A_1 , i.e. $B \times A_1$. Pasting those two pullbacks produces this square.

§3. $(B \times -)$ preserves covers

★ Punchline

In a pre-regular category, covers are stable under pullback. The square above exhibits $1 \times x$ as the pullback of x along π . Therefore: if $x : A_1 \rightarrow A_2$ is a cover, then $1 \times x : B \times A_1 \rightarrow B \times A_2$ is a cover (it is the pullback of a cover). So the functor $(B \times -)$ preserves covers, and hence $\Delta : \mathbf{A} \rightarrow \mathbf{A}/\mathbf{B}$, defined by $X \mapsto (X \times B \xrightarrow{\text{snd}} B)$, preserves covers too.